Package 'RegCalib'

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Type: Package

Version: 0.1.0

Title: Regression Calibration for Measurement Error Correction

License: GPL-3

Imports: stats, dplyr, magrittr, Matrix, matrixcalc, earth, survival

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URL: https://github.com/cuijy1990/RegCalib

BugReports: https://github.com/cuijy1990/RegCalib/issues

Description: Corrects for measurement error in continuous exposures (and covariates) using regression calibration methods. Provides corrected coefficients, standard errors, p-values, and variance-covariance matrices for linear, generalized linear, and Cox proportional hazards outcome models under both external and internal validation study designs. Implements the deattenuation factor method (RegCalibDF) and the substitution approach (RegCalibSub). Supports single and multiple mismeasured exposures.

Contents

RegCalibDF

Regression Calibration Using the Deattenuation Factor Method

Description

Corrects for measurement error in continuous exposures (and covariates) and returns corrected coefficients, standard errors, p-values, and variance-covariance matrices. Users may supply their own uncorrected estimates (e.g., from Cox or logistic models) instead of using the built-in outcome model. Both external and internal validation study designs are supported. Based on Rosner, Spiegelman & Willett (1989, 1990) and Spiegelman, McDermott & Rosner (1997).

Usage

```
RegCalibDF(
  supplyEstimates = FALSE, ms, vs,
  sur, exp, covCalib = NULL, covOutcomePlus = NULL,
  outcome = NA, event, time,
  method = "lm", family = NA, link = NA,
  external = TRUE,
  pointEstimates = NA, vcovEstimates = NA
)
```

Arguments

supplyEstimates
: Logical. If TRUE, uncorrected estimates are supplied by the user via pointEstimates and vcovEstimates, and ms is optional. Standard regression results are not returned. Default FALSE.

ms
: Main study data frame. Must contain all variables specified in sur, covCalib, and covOutcomePlus. Required when supplyEstimates = FALSE.

vs
: Internal or external validation study data frame. Must contain all variables specified in exp, sur, and covCalib.

sur
: Character vector of mismeasured exposure(s)/covariate(s) (surrogates) in the main study dataset.

exp
: Character vector of correctly-measured exposure(s)/covariate(s) in the validation dataset. Must correspond one-to-one with sur.

covCalib
: Character vector of correctly-measured covariates to adjust for in both the calibration model and outcome model. Default NULL.

covOutcomePlus
: Character vector of correctly-measured risk factors for the outcome not associated with the exposure or surrogate. Included in outcome model only; must not overlap with covCalib. Default NULL.

outcome
: Character. Name of the outcome variable. Required when method is "lm" or "glm".

event
: Character. Name of the event status indicator for Cox models (0 = censored, 1 = event). Required when method = "cox".

time
: Character. Name of the follow-up time variable for Cox models. Required when method = "cox".

method
: Character. Outcome modelling method: "lm", "glm", or "cox". Default "lm".

family
: Family function for glm (e.g., binomial). Not a character string. Required when method = "glm".

link
: Character. Link function for glm (e.g., "logit"). Required when method = "glm".

external
: Logical. TRUE (default) for external validation study; FALSE for internal. When FALSE, vs must contain the outcome variable.

pointEstimates
: Named numeric vector of uncorrected point estimates (intercept excluded). Required when supplyEstimates = TRUE.

vcovEstimates
: Named square matrix of uncorrected variance-covariance estimates (intercept excluded). Required when supplyEstimates = TRUE.

Value

A named list containing:

correctedCoefTable
: Data frame of corrected estimates, standard errors, Z-values, p-values, and 95% confidence intervals.

correctedVCOV
: Variance-covariance matrix of the corrected estimates.

standardCoefTable
: (When supplyEstimates = FALSE) Results from the uncorrected standard regression.

standardVCOV
: (When supplyEstimates = FALSE) Variance-covariance matrix from the uncorrected regression.

calibrationModelCoefTable
: Calibration model slope estimates.

calibrationModelVCOV
: Residual variance-covariance matrix from the calibration model.

Examples

```
result <- RegCalibDF(
  ms             = main_data,
  vs             = validation_data,
```

```
  sur           = "Z",
  exp           = "X",
  covCalib      = c("V1", "V2"),
  outcome       = "Y",
  method        = "lm",
  external      = TRUE
)
result$correctedCoefTable
```

References

Rosner B, Willett WC, Spiegelman D (1989). Correction of logistic relative risk estimates and confidence intervals for systematic within-person measurement error. *Statistics in Medicine* 8:1051-1069.

Rosner B, Spiegelman D, Willett WC (1990). Correction of logistic regression relative risk estimates and confidence intervals for measurement error: the case of multiple covariates measured with error. *American Journal of Epidemiology* 132:734-735.

Spiegelman D, McDermott A, Rosner B (1997). The many uses of the regression calibration method for measurement error bias correction in nutritional epidemiology. *American Journal of Clinical Nutrition* 65:1179S-1186S.

Spiegelman D, Carroll RJ, Kipnis V (2001). Efficient regression calibration for logistic regression in main study/internal validation study designs with an imperfect reference instrument. *Statistics in Medicine* 20:139-160.

See Also

RegCalibSub for the substitution approach.

RegCalibSub

Regression Calibration Using Substitution Method

Description

Corrects for measurement error in continuous exposures (and covariates) and returns corrected coefficients, standard errors, p-values, and variance-covariance matrices using the Carroll-Ruppert-Stefanski-Crainiceanu (CRS) sandwich variance estimator. Supports linear and generalized linear outcome models under external and internal validation study designs. Standard errors are derived analytically via the sandwich estimator (not bootstrap). Non-linear terms such as interaction terms should be pre-computed as permanent columns in the input datasets.

Usage

```
RegCalibSub(
  ms, vs, sur, exp, vsIndicator,
  covCalib = NULL, covOutcome = NULL,
  outcome = NA,
  method = "lm", family = NA, link = NA,
  external = TRUE
)
```

Arguments

`ms`
: Main study data frame. Must contain all variables specified in sur, outcome, and covOutcome (if any).

`vs`
: External validation study data frame. Must contain all variables specified in exp, sur, and covCalib (if any). Required when external = TRUE.

`sur`
: Character vector of mismeasured exposure(s)/covariate(s) (surrogates) in the main study dataset.

`exp`
: Character vector of correctly-measured exposure(s)/covariate(s) in the validation dataset. Must correspond one-to-one with sur.

`vsIndicator`
: Character. Name of the indicator variable in the main study identifying subjects with a validation record (1 = has validation record). Required when external = FALSE.

`covCalib`
: Character vector of correctly-measured covariates to adjust for in the calibration model, including any non-linear terms. Default NULL.

`covOutcome`
: Character vector of correctly-measured covariates to adjust for in the outcome model (with corrected exposure). Default NULL.

`outcome`

Character. Name of the outcome variable. Required.

method
: Character. Outcome modelling method: "lm" or "glm". Default "lm".

family
: Family function for glm (e.g., binomial). Not a character string. Required when method = "glm".

link
: Character. Link function for glm (e.g., "logit" or "log"). Required when method = "glm".

external
: Logical. TRUE (default) for an external validation study; FALSE for an internal validation study embedded in ms.

Value

A named list containing:

correctedCoefTable
: Matrix of corrected estimates, standard errors, Z-values, p-values, and 95% confidence intervals.

correctedVCOV
: Variance-covariance matrix of the corrected estimates.

Examples

```
# External validation study, linear model
result <- RegCalibSub(
  ms       = main_data,
  vs       = validation_data,
  sur      = "Z",
  exp      = "X",
  covCalib = c("V1", "V2"),
  outcome  = "Y",
  method   = "lm",
  external = TRUE
)
result$correctedCoefTable

# External validation, logistic model
result <- RegCalibSub(
  ms       = main_data,
  vs       = validation_data,
  sur      = "Z", exp = "X",
  covCalib = c("V1", "V2"),
  outcome  = "Y", method = "glm",
  family   = binomial, link = "logit",
  external = TRUE
)

# Internal validation study
result <- RegCalibSub(
  ms          = main_data,
  sur         = "Z", exp = "X",
```

```
  vsIndicator = "is_validated",
  covCalib    = c("V1", "V2"),
  outcome     = "Y", method = "lm",
  external    = FALSE
)
```

References

Carroll RJ, Ruppert D, Stefanski LA, Crainiceanu CM (2006). *Measurement Error in Nonlinear Models*, 2nd ed. Chapman & Hall/CRC.

See Also

RegCalibDF for the deattenuation factor method.